II PUC LBA-1 MODEL PAPER

SUBJECT : CHEMISTRY

UNIT-1: SOLUTIONS

Max : 30Marks

Instructions

1. Question paper has FIVE parts. All parts are compulsory.

2. a. Part-A carries 10 marks. Each question carries 1 mark.

b. Part-B carries 2 marks. Each question carries 2 marks.

c. Part-C carries 09 marks. Each question carries 3 marks.

d. Part-D carries 09marks. Each question carries 3 marks.

3. In Part-A questions, first attempted answer will be considered for awarding marks.

4. Write balanced chemical equations and draw neat labeled diagrams and graphs wherever necessary.

5. Direct answers to the numerical problems without detailed steps and specific unit for final answer will not carry any marks.

6. Use log tables and simple calculator if necessary (use of scientific calculator is not allowed).

PART-A

I. Select the correct option from the given choices. **7x1= 7**

1. Amalgam of mercury with sodium is an example for
   a) Solid solution with solute is liquid and solvent is solid
   b) Solid solution with solute is solid and solvent is liquid
   c) liquid solution with solute is solid and solvent is liquid
   d) liquid solution with solute is liquid and solvent is solid

2. A binary solution has two components 'A' and 'B'. The mole fraction of component 'A' is 0.5, then the number of moles of components 'A and 'B' in the solution is
   (a) $n_A > n_B$ (b) $n_A < n_B$ (c) $n_A = n_B$ (d) All of these

3. According to Dalton's law of partial pressures, the total pressure (p_{total}) over the solution phase in the container will be
   a) The product of the partial pressures of the components of the solution.
   b) The sum of the partial pressures of the components of the solution.
   c) The ratio of the partial pressures of the components of the solution.
   d) The difference of the partial pressures of the components of the solution.

4. The incorrect statement regarding the solution which is in dynamic equilibrium with undissolved solute is
   a) It is a saturated solution.
   b) It contains the maximum amount of solute dissolved in a given amount of solvent.
   c) the concentration of solute in such a solution is its solubility.
   d) It contains the minimum amount of solute dissolved in a given amount of solvent.

5. Which of the following aqueous solution has highest freezing point.
   a) 0.1mKCl (i = 2) b) 0.1mK_2SO_4 (i = 3)
   c) 0.1mAl_2SO_4 (i = 2) c) 0.1m sucrose (i =1).

6. A student took two glasses of pure water from a water filter. He cools glassA in a fridge and warms the other glass B on a stove. On comparing the solubility of oxygen in H_2O in glass–A and glass–B, he state that glass-'A' contains
   (a) more oxygen than glass–B. (b) less oxygen than glass–B.
   (c) same amount of oxygen as in glass–B. (d) zero concentration of oxygen.

7. **Statement–1:** If on mixing the two liquids, the solution becomes hot, it implies that it shows negative deviation from Raoult's law.
   Statement–2: Solutions which show negative deviation are accompanied by decrease in volume.
   (a) Both Statement–1 and Statement–2 are true.
   (b) Statement–1 is true but Statement–2 is false;
   (c) Both Statement–1 and Statement–2 are false.
   (d) Statement–1 is false but Statement–2 is true

II. Fill in the blanks by choosing the appropriate word from those given in the brackets:
[ppm. Increases, 95% decreases] **1 × 3 = 3**

8. Fractional distillation of mixture of fermentation of sugars gives a solution containing approximately _____of ethanol.
9. The concentration term used to express when solute is present in trace quantities is _____
10. As pressure increases solubility of gas in liquid_____

PART - B

III. Answer any one of the following. Each question carries 2 marks. **1 × 2 = 2**

11. State Henry's Law. Give any one application of it.
12. Define isotonic solutions. What happens to flow of solvent molecules in isotonic solutions?

PART – C

IV. Answer any three of the following. Each question carries 3 marks **3 × 3 = 9**

13. Write three reasons to justify that osmotic pressure method has the advantage over other colligative methods for the measurement of molar mass of the macromolecules.
14. Define azeotropic mixtures. Which type of nonideal solution shows maximum boiling azeotropic solution? Give one example for it.
15. Write any three differences between ideal and non-ideal solutions.
16. Name the concentration term which is commonly used in medicine and pharmacy. Write the definition and mathematical equation for that concentration term.
17. Name any three factors affecting solubility of solute in solvent.

PART – D (PROBLEMS)

V. Answer any three of the following. Each question carries 3 marks **3 × 3 = 9**

18. The boiling point of benzene is 353.23K. Calculate the mass of non-volatile solute is to be added to 90g of benzene such that it boils at 354.11K. The molar mass of solute is 58 $gmol^{-1}$ [Given: Kb = 2.53 K Kg mol^{-1}].
19. Calculate the mass of a non-volatile solute which should be dissolved in 114g octane to reduce its vapour pressure to 80%. [Given Molecular formula of octane is C8H18; Molar mass of solute = 40 $gmol^{-1}$].
20. Henry's law constant for CO_2 in water is 1.67×10^8 Pa at 298 K. Calculate the quantity of CO_2 in 500 mL of soda water when packed under 2.5 atm CO_2 pressure at 298 K.
21. Calculate the osmotic pressure of 0.5%(w/v) aqueous solution sucrose at 300K. (R = 0.0083 L bar mol^{-1} and molar mass of sucrose = 342 g mol^{-1}).
22. The vapour pressure of pure liquids A and B are 450 and 700 mm Hg respectively, at 350 K . Find out the composition of the liquid mixture if total vapour pressure is 600 mm Hg. Also find the composition of the vapour phase.
23. 0.6 mL of acetic acid (CH_3COOH), having density 1.06 gcm^{-3}, is dissolved in 1 litre of water. The depression in freezing point observed for this strength of acid was 0.0205°C. Calculate the van't Hoff factor.

BLUE PRINT OF LBA -1 MODEL QUESTION PAPER:

UNIT: SOLUTIONS

UNIT NUMBER	UNIT NAME	TEACHING HOURS	NUMBER OF MARKS	PART-A	PART-B	PART-C	PART-D
				1 MARK	2 MARK	3 MARK	3 MARK
1	SOLUTIONS	14	30	**10**	**2**	**9**	**9**
TOTAL	NUMBER OF QUESTION			**10/10**	**1/2**	**3/5**	**3/6**
	NUMBER OF MARKS			**10/10**	**2/4**	**9/15**	**9/18**